

ANIRUDH ARORA

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RESEARCH INTERESTS

Tropical precipitation and Indian Summer Monsoon dynamics; evaluation of CMIP-class and kilometre-scale Earth system models; large ensembles and quantification of internal variability; atmospheric composition and satellite-based trend detection of greenhouse gases; climate model uncertainty and its translation into actionable, policy-relevant science.

EDUCATION

Indian Institute of Science Education and Research, Bhopal <i>Bachelor and Master of Science in Environmental Sciences, CPI : 9.15/10.00</i>	2021 - 2026 <i>Bhopal, MP</i>
St. Paul H.S. School <i>High School STEM Diploma, 10th (CBSE): 95.2%, 12th (CBSE): 95.6%</i>	2006 - 2021 <i>Indore, MP</i>

FORMAL RESEARCH EXPERIENCE

Max Planck Institute for Meteorology (MPI-M) – <i>incoming</i> <i>July 2026 – November 2026</i>	Climate Physics Department <i>Hamburg, Germany</i>
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- Five-month visiting scientist position in the Climate Physics Department, working on cloud microphysics.

CLIMATics Group, KAUST <i>July 2025 - Present</i>	Supervisor - Prof. Ibrahim Hoteit <i>Thuwal, Saudi Arabia</i>
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- Master's thesis: *Toward Reliable Tropical Precipitation Simulations by Assessing the Added Value of Large Ensembles and Kilometer-Scale Simulations.*
- Evaluating the IFS-FESOM nextGEMS production historical simulation at kilometre scale against IMD, GPCP, ERA5 and CMIP6 MMM benchmarks; produced a manuscript which is under review.
- Quantifying ensemble convergence (N_{eff}) for CESM2-LE, MPI-GE and CESM2-PPE precipitation using four criteria, including a probabilistic Brier Skill Score formulation, to assess the sample size needed to reliably constrain monsoon mean state and intraseasonal variability.
- *Methane Variability and Trend Analysis over the Arabian Peninsula*: Conducted the first dedicated long-term satellite-based analysis of atmospheric methane over the Arabian Peninsula using 93 months (May 2018 – December 2025) of bias-corrected blended TROPOMI+GOSAT XCH₄ retrievals. Manuscript in preparation for submission to *Atmospheric Chemistry and Physics* (ACP)

DST CoE for Climate Information, IIT Delhi <i>May 2023 - Present</i>	Supervisor - Prof. Saroj K. Mishra <i>New Delhi, India</i>
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- Co-supervised by Prof. Mishra for my master's thesis.
- Evaluated CMIP5/CMIP6 multi-model ensemble and IFS-FESOM km-scale simulations of historical precipitation and Indian Summer Monsoon over the South Asian domain; manuscript in preparation.
- Investigated circulation-driven precipitation and temperature responses under solar geoengineering (GeoMIP G1 and G6sulfur), characterising regional monsoon and tropical rainfall changes.
- Co-developed a sequential decision-making (MDP/RL) framework for efficient climate action; currently part of a manuscript in preparation.
- Contributed to a multi-scale source attribution of natural and anthropogenic air pollution over the Indo-Gangetic Plain using reanalysis and satellite datasets, published as a perspective in *The India Forum*.
- Co-author on the SARCI framework for climate services in South Asia, accepted at the *Bulletin of the American Meteorological Society*.

August 2023 - April 2024

- Conducted a field ethnography study to understand the socio-cultural importance of Garuda Purana, the book about death rituals and life after death.
- Conducted a comprehensive ethnographic study of two distinct laboratory settings over a four-month period, aiming to understand the socio-cultural dynamics, work practices, and interactions within these scientific environments.

PUBLICATIONS AND CONFERENCE TALKS

- Arora, A.**, Dasari, H.P., Jha, S.K., Mishra, S.K., Hoteit, I. (2026), *Evaluating Indian Summer Monsoon Precipitation in the IFS-FESOM nextGEMS Production Historical Simulation at Kilometre Scale*. Accepted at the km-Scale Global Modelling Summit 2026, Hamburg, Germany.
- Bhuyan, D. P., Upadhyaya, P., Pathak, R., Namdev, P., Salunke, P., Anand, A., Suresh, A. D., **Arora, A.**, Baraik, S. K., Jain, S., Parihar, R. S., Dwivedi, A., Sahany, S., Sharan, M., Dash, S. K., Fasullo, J. T., Behera, S. K., Tribbia, J. and Mishra, S. K. (2026), *Advancing climate services in South Asia: The SARCI Framework for actionable information, capability, and capacity*. Bulletin of the American Meteorological Society, BAMS-D (DOI 10.1175/BAMS-D-25-0172.1).
- Mishra, S. K., Upadhyaya P., **Arora A.**, Vinod A.J., Jayan N., Dwivedi A. (2025), *To Solve North India's Air Crisis, Look Beyond Stubble and Fireworks*. The India Forum, **Oct 22, 2025**.
- Arora, A.**, Upadhyaya, P., Mishra, S.K. (2024), *Circulation Changes under Geoengineering and their Implications in Regional Precipitation and Temperature*. 14th Annual GeoMIP Meeting 2024, Cornell University, USA. (Poster)

MANUSCRIPT UNDER PREPARATION

- Arora, A.**, Gandhi, H., Hoteit, I. (2026), *Methane variability, persistent hotspots, and shamal-driven seasonality over the Arabian Peninsula from blended TROPOMI+GOSAT XCH₄ retrievals (2018–2025)*. In preparation for *Atmospheric Chemistry and Physics*.
- Arora, A.**, Dasari, H.P., Jha, S.K., Mishra, S.K., Hoteit, I. (2026), *Evaluating Indian Summer Monsoon Precipitation in the IFS-FESOM nextGEMS Historical Simulation at Kilometre Scale*.
- Arora, A.** & Upadhyaya, P., Mishra, S.K., Bhuyan, D., Behera, S., Fasullo, S., Tribbia, J., Jain, S., Salunke, P., Sahany, S., Dash, S.K. (2026), *Climate models doubled in a decade but credibility remains poor over the Global South*.
- Arora, A.** (2026), *Public enlightenment and climate action: rethinking Kantian principles for a sustainable future* (under review at *npj Climate Action*).

TECHNICAL SKILLS

Languages	Python, MATLAB, NCAR Command Language (NCL), Mathematica, SQL
Climate tools	Climate Data Operators (CDO), netCDF Operators (NCO), NetCDF
HPC & workflows	Slurm, Levante (DKRZ), Shaheen III (KAUST), Bash, Git
Typesetting & Web	L ^A T _E X, HTML5, CSS

OTHER RELEVANT PROJECTS

Modeling Atmospheric Aerosols in Climate Systems Advisor - Prof. Ramya Sunder Raman
August 2024 - November 2024 IISER Bhopal

- A comprehensive report on understanding aerosol dynamics and their representation in GCMs, focusing on aerosol microphysics and their interactions with clouds and radiative forcing.

Aerosols and Their Influence on Indian Monsoon Rainfall Advisor - Dr. Shubhi Agrawal
August 2024 - November 2024 IISER Bhopal

- Co-authored a review term paper as part of the course "Indian Monsoon and its Variability" analyzing the influence of remote aerosol emissions on South Asian monsoon variability.

Modelling the Perfect Penalty kick

November 2022

Independent

IISER Bhopal

- Obtained initial velocities and spin values that must be imparted to a football for a perfect penalty kick, by computationally solving the six coupled differential equations we obtained with varying initial conditions. It can be viewed [here](#)

HONORS

Rhodes Scholarship 2026 Finalist

University of Oxford – Rhodes Trust

November 2025

- Among the top 13 candidates out of ~1,000 applicants for the Rhodes Scholarship India.

Visiting Student Research Program

King Abdullah University of Science and Technology (KAUST)

March 2025

Kingdom of Saudi Arabia

- Received full scholarship to conduct master's thesis research at KAUST under Prof. Ibrahim Hoteit.

MPI-M 1-year Research Funding (Declined)

Max Planck Institute for Meteorology

March 2025

Hamburg, Germany

- Received a full funding to conduct research at MPI-M, Germany as part of IISER-MPI collaboration.

Full Travel Support – 14th Annual GeoMIP Meeting

Cornell University

July 2024

Ithaca, NY, USA

- Full travel and accommodation support to present at the 14th Annual GeoMIP Meeting.

CNR Rao Education Foundation Prize

Indian Institute of Science Education and Research

August 2022

Bhopal, MP

- Awarded for the highest CPI (10.00/10.00) in the first semester of the BS-MS dual-degree programme.

Bronze Medal – The University Physics Competition

Carroll College, Helena, MT

January 2023

- Team award in the worldwide University Physics Competition 2022.

Earlier honours

- GIZ Sustainable Development Competition (felicitated at the German Embassy, New Delhi, 2019); Joseph Memorial Scholarship for academic excellence and leadership, St. Paul H.S. School, Indore (2019).

POSITIONS OF RESPONSIBILITIES

Prayaas - Making Lives Colorful

Coordinator

Student Organization

2022 - 2026

- Prayaas is a volunteer group working towards the upliftment of underprivileged communities. I actively contributed to community empowerment through education and social initiatives.

IISER Bhopal Debating Society

President

Student Organization

2022

Chrysalis IISER Bhopal

Podcast Creator

Science Communication Body

2022

- Co-hosted a podcast on the topic, "Climate Change Research Governing Public Policy Making" and talked about the challenges and opportunities to integrate climate change research with public policymaking.